MODEL CASE STUDIES — LAB INTERNAL

CASE STUDY 1: Student Record Management System

Main Page

Develop a student management application with options to add, view, search, update and delete student records.

(a) Add Student

- **Collect Student ID, Name, Roll Number, Branch, Year, Email and Phone Number.**
- **Validate mandatory fields and insert the student details into the database.**

(b) View/Search Students

- **Retrieve and display all student records in a table.**
- **Allow searching by Student ID, Roll Number or Name.**

(c) Update Student Details

- **Retrieve the existing details using Student ID.**
- **Allow modification of branch, year, email and phone number.**
- **Save the updated details in the database.**

(d) Delete Student

- **Delete a student record using Student ID.**
- **Display a confirmation message and handle records that do not exist.**

(e) Mandatory MERN Module

Implement at least one student database interaction using the MERN stack, such as adding a student through a React form or retrieving student records from MongoDB and displaying them in a React table.

CASE STUDY 2: Library Book Management System

Main Page

Develop a library application with options to add, view, search, update and delete book records.

(a) Add Book

- **Collect Book ID, Title, Author, Category, ISBN and Number of Copies.**

- **Validate the input and save book details.**

(b) View/Search Books

- **Display all books in a table.**
- **Allow searching by Book ID, Title or Author.**
- **Display the availability status and number of copies.**

(c) Update Book

- **Retrieve book details using Book ID.**
- **Allow modification of title, author, category and number of copies.**
- **Save the changes.**

(d) Delete Book

- **Delete a book record using Book ID.**
- **Check whether the book exists before deletion.**

(e) Mandatory MERN Module

Implement at least one book database operation using React.js, Node.js, Express.js and MongoDB.

CASE STUDY 3: Employee Information Management System

Main Page

Create an employee management dashboard with Add Employee, View Employees, Update Employee and Delete Employee options.

(a) Add Employee

- **Collect Employee ID, Name, Department, Designation, Salary, Email and Joining Date.**
- **Validate the fields and insert the record.**

(b) View Employees

- **Retrieve and display all employee records.**
- **Allow searching by Employee ID and filtering by department.**

(c) Update Employee

- **Retrieve an employee record using Employee ID.**
- **Allow modification of department, designation, salary and contact details.**

(d) Delete Employee

- **Delete an employee record using Employee ID.**
- **Display suitable success or error messages.**

(e) Mandatory MERN Module

Implement at least one employee database interaction using the MERN stack.

CASE STUDY 4: Product Inventory Management System

Main Page

Develop an inventory application with Add Product, View Products, Update Product and Delete Product options.

(a) Add Product

- **Collect Product ID, Product Name, Category, Price and Quantity.**
- **Validate the details and save them in the database.**

(b) View/Search Products

- **Display all product records.**
- **Allow searching by Product ID or Product Name.**
- **Display the available stock.**

(c) Update Product

- **Allow modification of product price, category and quantity.**
- **Update the corresponding database record.**

(d) Delete Product

- **Delete a product using Product ID.**
- **Display a suitable message if the product does not exist.**

(e) Mandatory MERN Module

Implement at least one inventory database operation using the MERN stack.

CASE STUDY 5: Course Registration Management System

Main Page

Develop a course management application with Add Course, View Courses, Update Course, Delete Course and Register for Course options.

(a) Add Course

- **Collect Course ID, Course Name, Faculty Name, Duration and Available Seats.**
- **Save the course information in the database.**

(b) View/Search Courses

- **Display all available courses.**

- **Allow searching by Course ID or Course Name.**

(c) Update Course

- **Retrieve a course using Course ID.**
- **Allow modification of course details and available seats.**

(d) Delete Course

- **Delete a course record using Course ID.**
- **Display a confirmation message.**

(e) Student Enrollment

- **Allow students to register for available courses.**
- **Prevent duplicate registrations and enrollment when seats are unavailable.**
- **Display registered courses.**

(f) Mandatory MERN Module

Implement at least one course or enrollment database interaction using the MERN stack.

CASE STUDY 6: Hospital Patient Record Management System

Main Page

Develop a patient management application with Add Patient, View Patients, Update Patient and Delete Patient options.

(a) Add Patient

- **Collect Patient ID, Patient Name, Age, Gender, Contact Number, Doctor Name and Appointment Date.**
- **Validate and store the patient details.**

(b) View/Search Patients

- **Display patient records in a table.**
- **Allow searching by Patient ID or Patient Name.**

(c) Update Patient

- **Retrieve patient details using Patient ID.**
- **Allow updating contact information, doctor name and appointment date.**

(d) Delete Patient

- **Delete a patient record using Patient ID.**
- **Display appropriate success or error messages.**

(e) Mandatory MERN Module

Implement at least one patient-record database interaction using the MERN stack, with suitable access restrictions.

CASE STUDY 7: Vehicle Service Booking System

Main Page

Develop a vehicle service application with Customer Registration, Vehicle Registration, Book Service, View Service History and Update Booking options.

(a) Register Vehicle

- **Collect Vehicle Registration Number, Make, Model and Manufacturing Year.**
- **Associate the vehicle with the customer.**

(b) Book Service

- **Collect Vehicle Number, Service Type, Preferred Date and Time Slot.**
- **Validate the booking details and save the booking.**

(c) View Service History

- **Display previous and upcoming service bookings.**
- **Show service date, vehicle details and booking status.**

(d) Update/Cancel Booking

- **Allow the customer to update eligible bookings or cancel them.**
- **Display the updated booking status.**

(e) Mandatory MERN Module

Implement at least one vehicle or service-booking database operation using the MERN stack.

CASE STUDY 8: Event Registration and Attendance Portal

Main Page

Develop an event management application with Add Event, View Events, Register for Event and View Registrations options.

(a) Add Event

- **Collect Event ID, Event Name, Date, Venue, Category and Maximum Participants.**
- **Store event details in the database.**

(b) View/Search Events

- **Display upcoming events.**

- **Allow filtering by event date or category.**

(c) Register for Event

- **Collect Participant ID, Participant Name and Event ID.**
- **Prevent duplicate registrations and registrations when seats are full.**

(d) Manage Registrations

- **Display registered participants.**
- **Allow the organizer to update registration status and record attendance.**

(e) Mandatory MERN Module

Implement at least one event or participant-registration database operation using the MERN stack.

CASE STUDY 9: Apartment Maintenance Request System

Main Page

Develop a maintenance application with Submit Request, View Requests, Update Request and Manage Requests options.

(a) Submit Maintenance Request

- **Collect Resident ID, Apartment Number, Issue Category, Description and Priority.**
- **Save the request with a Pending status.**

(b) View Requests

- **Display requests submitted by the resident.**
- **Show request date, category, priority and status.**

(c) Update Request

- **Allow eligible requests to be updated with additional details.**
- **Save the modified information.**

(d) Manage Requests

- **Allow the manager to view all requests.**
- **Update the status and add remarks.**

(e) Mandatory MERN Module

Implement at least one maintenance-request database operation using the MERN stack.

CASE STUDY 10: Lost and Found Campus Portal

Main Page

Develop a campus portal with Report Item, Search Items, Update Report and Delete/Resolve Report options.

(a) Report Lost/Found Item

- **Collect Item Name, Description, Category, Location, Date and Report Type (Lost or Found).**
- **Store the report with an Open status.**

(b) Search Item Reports

- **Display matching reports.**
- **Allow searching by item name, category, location or report type.**

(c) Update Report

- **Allow the user to edit their own item report.**
- **Save the modified details.**

(d) Delete/Resolve Report

- **Allow the user to delete their own report or mark it as Resolved.**
- **Display an appropriate status message.**

(e) Mandatory MERN Module

Implement at least one lost-and-found database operation using the MERN stack.

CASE STUDY 11: Laboratory Equipment Issue Register

Main Page

Develop an equipment management application with Add Equipment, View Equipment, Issue Equipment, Record Return and Update Equipment options.

(a) Add Equipment

- **Collect Equipment ID, Equipment Name, Category and Available Quantity.**
- **Store the equipment details.**

(b) View/Search Equipment

- **Display equipment records.**
- **Allow searching by Equipment ID or Equipment Name.**

(c) Issue Equipment

- **Collect Student Roll Number, Student Name, Equipment ID, Issue Date and Due Date.**
- **Allow issuing only when sufficient quantity is available.**

- **Update the available quantity.**

(d) Record Return

- **Collect return date and equipment condition.**
- **Update the issue record and available quantity appropriately.**

(e) Mandatory MERN Module

Implement at least one equipment or issue/return database operation using the MERN stack.

CASE STUDY 12: Internship Application Tracker

Main Page

Develop an internship portal with Add Internship, View Internships, Apply for Internship and Manage Applications options.

(a) Add Internship

- **Collect Internship ID, Organization, Role, Eligibility, Location, Stipend and Last Date.**
- **Store internship details in the database.**

(b) View/Search Internships

- **Display available internship opportunities.**
- **Allow searching by organization, role or location.**

(c) Apply for Internship

- **Collect Student Roll Number and Internship ID.**
- **Store the application date and initial status.**
- **Prevent duplicate applications and applications submitted after the deadline.**

(d) Manage Applications

- **Display applicants for each internship.**
- **Allow the coordinator to update application status to Shortlisted, Rejected or Selected.**

(e) Mandatory MERN Module

Implement at least one internship or application database operation using the MERN stack.